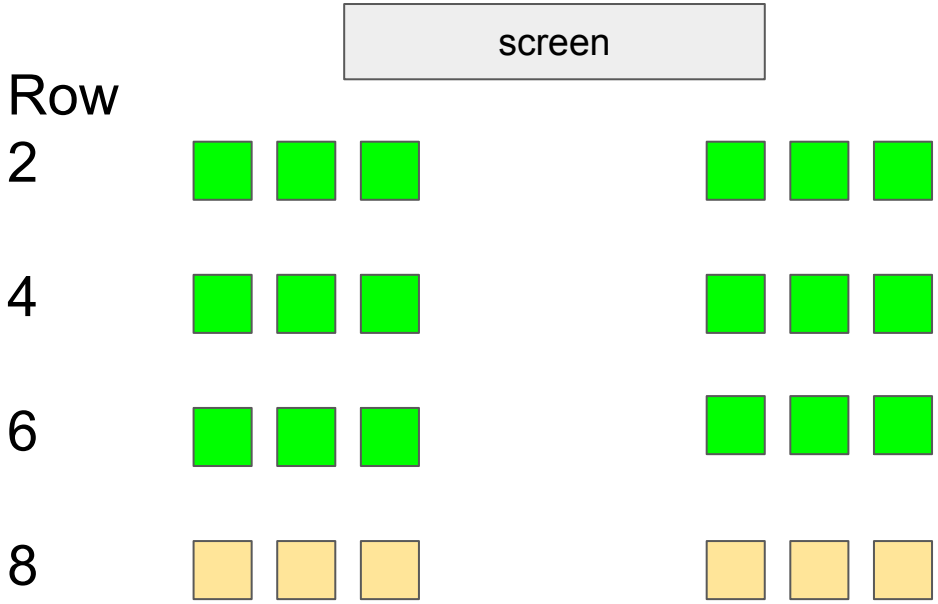
Please do NOT sit in the 1st, 3rd, or 5th rows!

Sit at a square. Trying to get some little groups so you all can have more support from each other while working through the notebook.



Yellow squares for those who prefer to work (almost) entirely independently

Lab 2

Census Blocks & the ACS

American Community Survey

Introduction to GEOIDs, power laws, box plots, & heatmaps

DATA 20195 Spring 2026

Emma Ross and Michael Xiao

Announcements

- Welcome back to lab!
- **Homework 2 due Wednesday, April 8th, 11:59 pm**, with automatic extension to Friday, April 10th, 1:30pm *(read syllabus for details)*
- Readings
  - **Week 2:** Ian Hacking, Kinds of People: Moving Targets
  - **Week 3:** Moon will send out an email ASAP to give reading assignments!
  - Printed copies available **next week in class**
  - Further optional readings available on Github
- Office Hours in Ryerson N375B
  - Emma: Tuesdays, 2–3pm
  - Michael: Thursdays, 2–3pm

Warm-Up Question

	Year	Candidate	Party	Popular vote	Result	%
0	1824	Andrew Jackson	Democratic-Republican	151271	loss	57.210122
1	1824	John Quincy Adams	Democratic-Republican	113142	win	42.789878
2	1828	Andrew Jackson	Democratic	642806	win	56.203927
3	1828	John Quincy Adams	National Republican	500897	loss	43.796073
4	1832	Andrew Jackson	Democratic	702735	win	54.574789
...	...	...	...	...	...	...
182	2024	Donald Trump	Republican	77303568	win	49.808629
183	2024	Kamala Harris	Democratic	75019230	loss	48.336772
184	2024	Jill Stein	Green	861155	loss	0.554864
185	2024	Robert Kennedy	Independent	756383	loss	0.487357
186	2024	Chase Oliver	Libertarian Party	650130	loss	0.418895

```
election_df
```

What does the following code output?

```
election_df[(election_df["Result"] == win) & (election_df["%"] >= 50)]
```

Example adapted from UC Berkeley's Data 100 (Principles and Techniques of Data Science) Summer 2025

From last time…

Tabular data in `pandas` is referred to as `DataFrame`

Access specific columns of a `DataFrame`:

`df[["column_name","column_name"]]` or `df.loc[[0,1,3]]`

Utility functions

`matplotlib` is a standard Python library for basic plotting

- Histograms
- Scatter plots
- Line graphs
- Pie chart

pandas (*not exhaustive)

`df.head(2)` prints first two rows of `Dataframe`

`df.sort_values(by="col_1", ascending=True)` sorts rows of `Dataframe` by values in `col_1`, lowest values first

`df.rename(columns= {"old_col_1": "new_col_1"} )` renames `old_col_1` to `new_col_1` using dictionary syntax

`df.set_index("id_or_name_col")` sets the `id_or_name_col` to be the index (row labels) of `Dataframe`

`df.drop(columns=["col_1", "col_2"])` removes `col_1` and `col_2` from `Dataframe`

`df["col_1"].sum()` sums down `col_1`

`df.sum(axis=0)` sums down all columns of `Dataframe`

`df["new_col"] = df.sum(axis=1)` appends a row sum column called `new_col` to the end of `Dataframe`

`df.loc[df["col_1"] > 15]` filters `Dataframe` to rows where the value of `col_1` is greater than 15

`df.loc[(df["col_1"] > 15) & (df["col_1"] < 70)]`

filters `Dataframe` to rows where the value of `col_1` is greater than 15 **AND** the value of `col_1` is less than 70

`df.loc[(df["col_1"] > 15) | (df["col_2"] > 20)]`

filters `Dataframe` to rows where the value of `col_1` is greater than 15 **OR** the value of `col_2` is greater than 20

matplotlib (*not exhaustive)

```
fig, ax = plt.subplots()
```

Line: `ax.plot(df["x_col"], df["y_col"])`

Scatter: `ax.scatter(df["x_col"], df["y_col"], s= size_pts, color="color_name_of_pts")`

Histogram:

```
ax.hist(df["col"], bins=np.arange(min_bin_value, max_bin_value, bin_increment))
ax.set_xticks(np.arange(min_tick_value, max_tick_value, tick_increment))
```

Pie plot: `df[["col_1", "col_2", "col_3"]].loc["row_name"].plot.pie(figsize=(10, 10), legend=True, ax=ax)`

Axis markings:

```
ax.set_xlabel("informative label based on x_col values")
ax.set_ylabel("informative label based on y_col values")
ax.set_xlim(x_axis_min, x_axis_max)
ax.set_ylim(y_axis_min, y_axis_max)
```

Graph configuration and saving:

```
ax.set_title("informative title")
ax.set_aspect("equal")
plt.savefig("name_to_save_as.png")
plt.show()
```

👀Make sure you pay attention to the order of the last two commands to avoid blank plots!

Census Hierarchy

Census Hierarchy

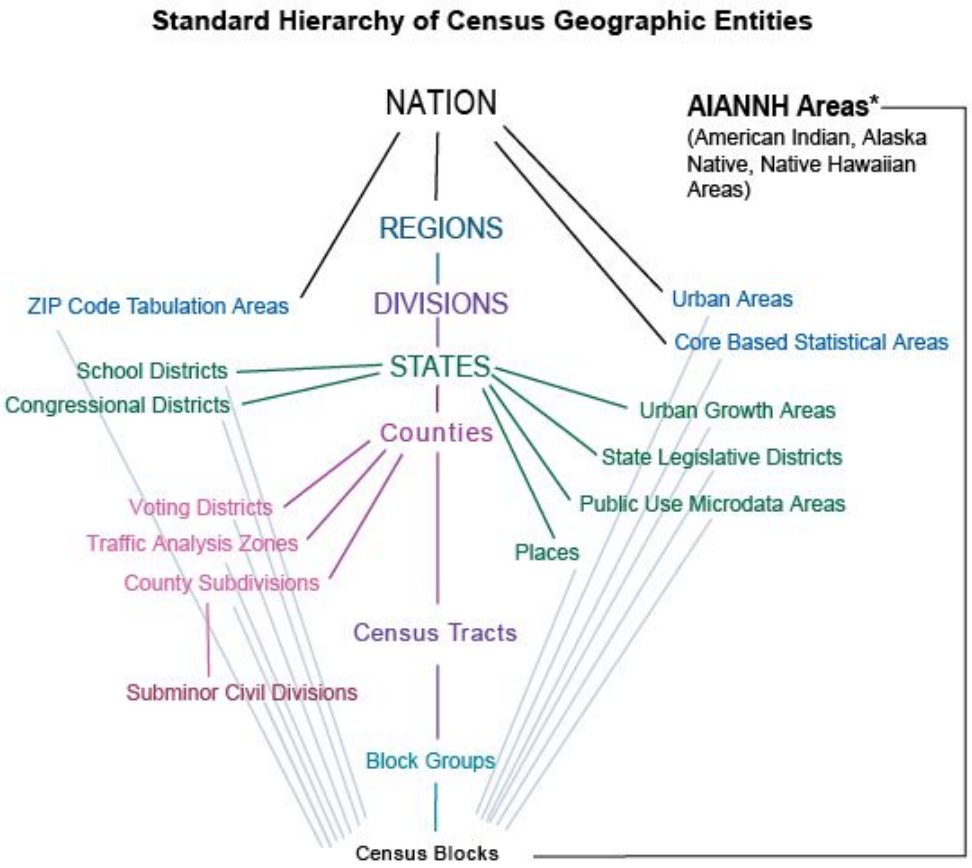


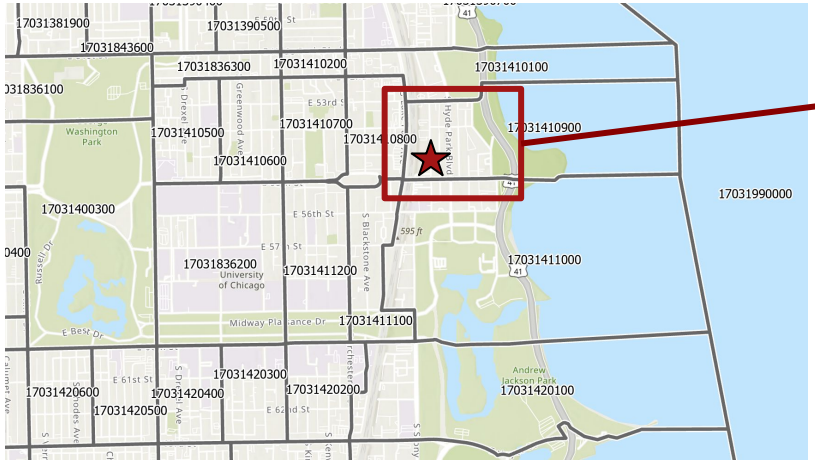
Image credit: University of Missouri

GEOIDs

GEOIDs are the numeric codes the Census uses to name their geographies, below are some examples:

17 031 410900 3 011

State FIPS | County | Tract | Block Group | Block



https://mtgis-portal.geo.census.gov/arcgis/apps/experiencebuilder/experience/?id=bc7d5cafd5e94dfb875ac36df0deaf77

https://tigerweb.geo.census.gov/tigerweb/

American Community Survey (ACS)

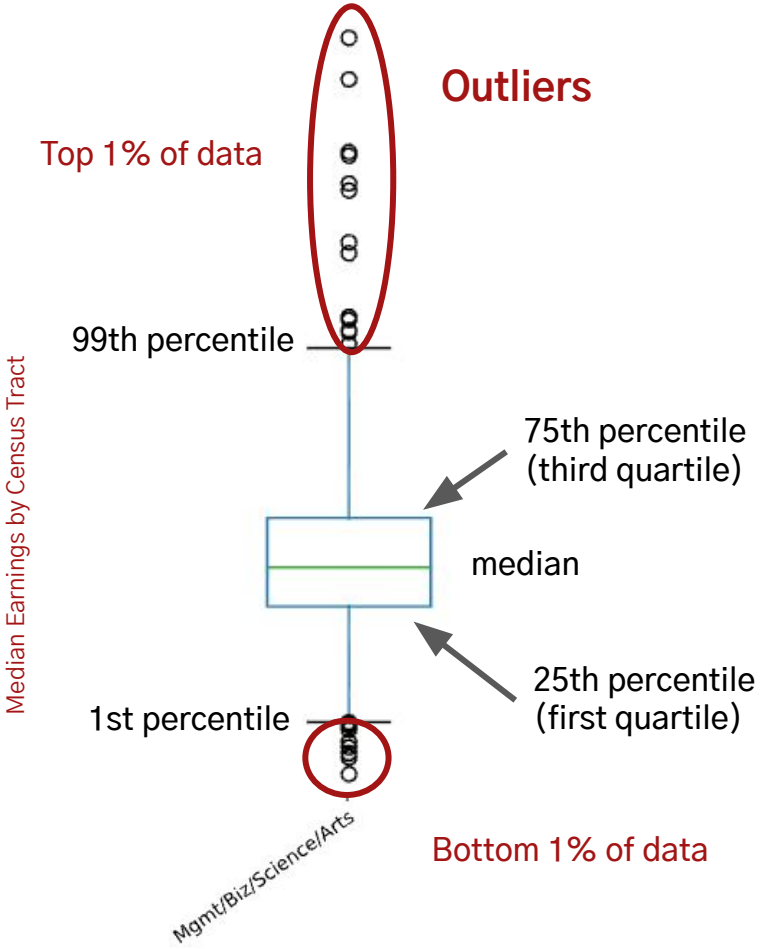
AMERICAN COMMUNITY SURVEY
U.S. CENSUS BUREAU

	Decennial Census	ACS 5-year estimates
Coverage	Full count	~3.5M households/year (sample)
Frequency	Every 10 years	Annually (pooled over 5 years)
Variables	Demographics only	Demographics + socioeconomics
Smallest unit	Block	Tract (usually)
Uncertainty	Exact counts	Has margins of error

Google sheets workbook that organizes this information for the 2010 and 2020 Decennial Census plus the ACS vintages from 2013 to 2023.

New `matplotlib` and `pandas` methods

Boxplot

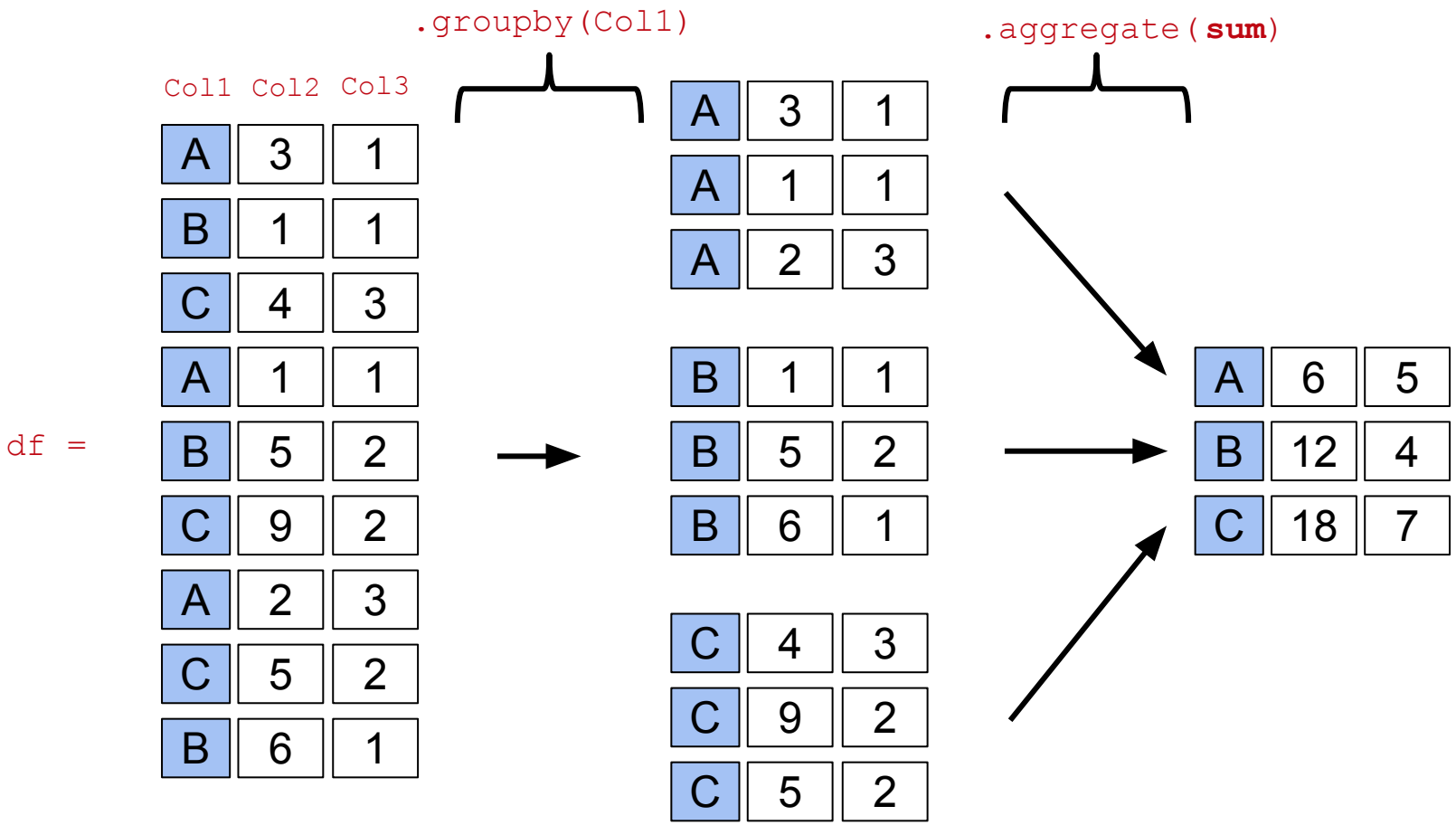



```
fig, ax = plt.subplots()
df.boxplot(
    ax=ax,
    rot=35,
    whis=(1,99)
)
```

rotate x-axis labels for readability

set whiskers to 1st and 99th percentiles

groupby



Demo adapted from UC Berkeley's Data 100 (Principles and Techniques of Data Science) Summer 2025

join

df1

ID		
A	6	ca
B	3	il
C	5	tx

df2

ID			
C	hi	1.1	2
A	hey	3.4	7
B	yay	2.5	9

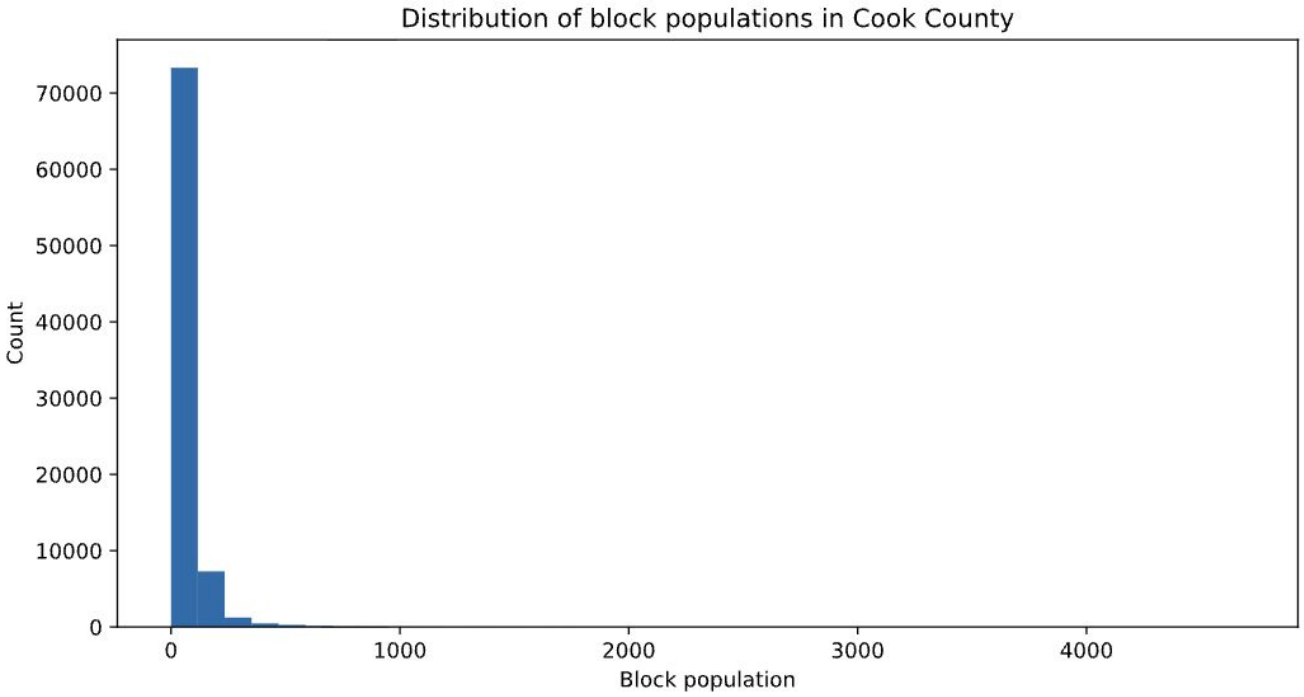
df1.join(df2)

A	6	ca	hey	3.4	7
B	3	il	yay	2.5	9
C	5	tx	hi	1.1	2

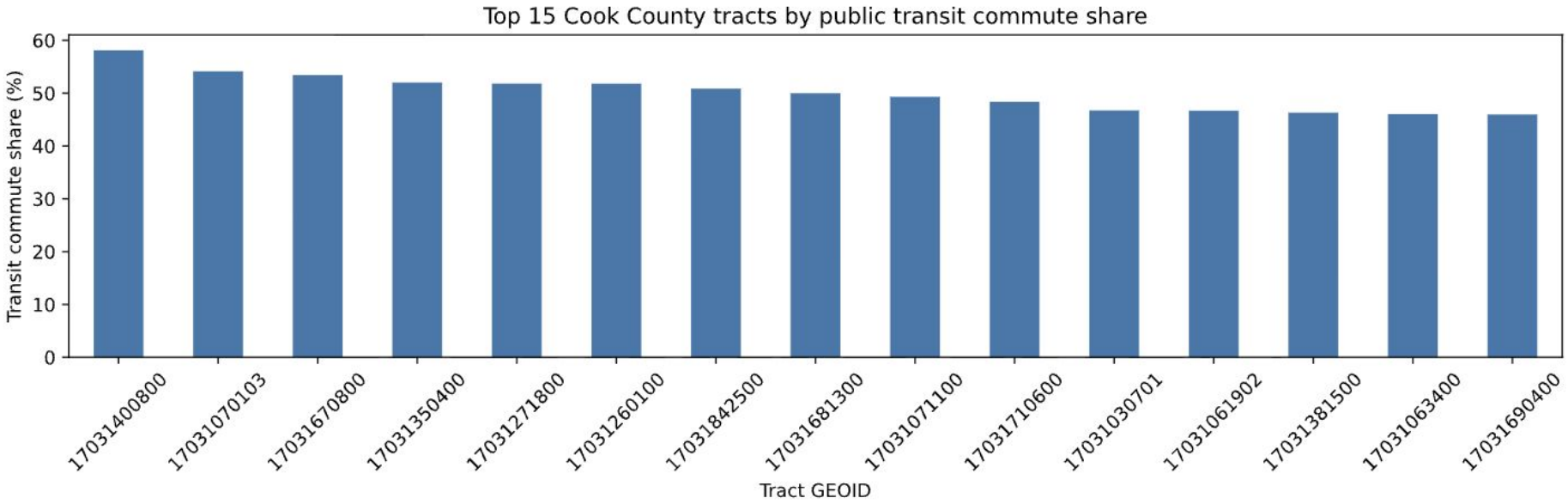
Demo adapted from UC Berkeley's Data 100 (Principles and Techniques of Data Science) Summer 2025

Visualization Clinic

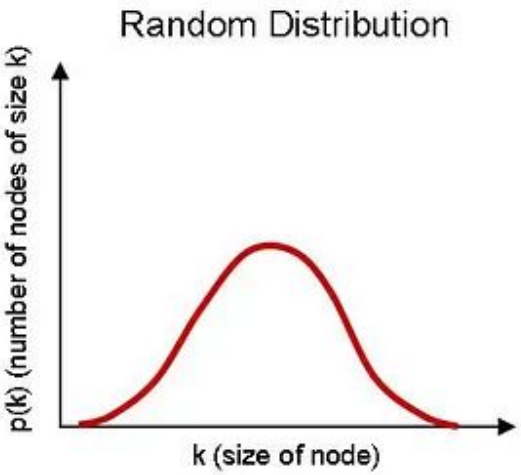
How can we improve this plot?

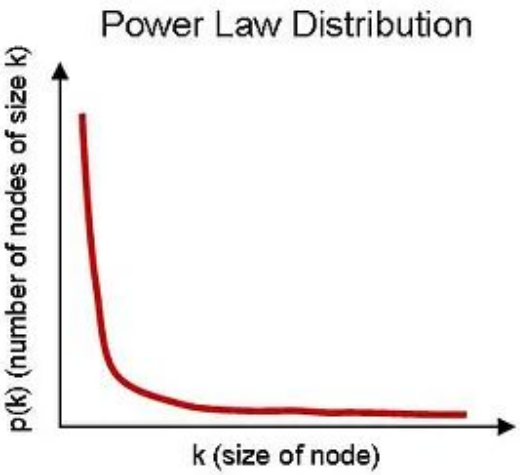


How can we improve this plot?



Powerlaws





https://medium.com/@jithamithra/the-power-law-or-why-working-hard-is-not-enough-b5d874195403

Heat maps

Illinois Counties: Median Earnings by Educational Attainment
ACS 5-Year (2023)
No degree
High school diploma
Associate's
Bachelor's
Master's
Graduate / Professional
Adams
Alexander
Bond
Boone
Brown
Bureau
Calhoun
Carroll
Cass
Champaign
90000
80000
70000
60000
50000
40000
30000
20000
Median Earnings ($)

https://tinyurl.com/20195-lab2